

NQ_TPO_VolumeMap v1.0

thinkorswim study: TPO (MonkeyBars) + volume profile for /NQ and /ES on a 30-minute chart

1. Install

1. Open thinkorswim → Charts → **Studies** → **Edit Studies** → **Create** (or open the existing TPO_VolumeMap study).
2. Paste the full contents of **NQ_TPO_VolumeMap_v1.0.txt**, replacing any older code. Name it and click **OK**.
3. Set the chart aggregation to **30 minutes**. Include extended hours (ETH) so Globex profiles are drawn.
4. If an older version was already on the chart, **remove the study and add it again** so old saved inputs are cleared.
5. Set **rowHeight** for the chart: **/NQ = 5** (default), **/ES = 2**, other symbols = 2.

2. Sessions and TPO periods

Profile	Time (US Eastern)	Period numbers
RTH	09:30 – 18:00 (includes the 16:00–17:00 post-close bars)	09:30 = 1 ... 15:30 = 13, 16:00 = 14, 16:30 = 15
Globex	18:00 – 09:30 (overnight, crosses midnight)	18:00 = 1, 00:00 = 13 ... 09:00 = 31

Digits show 1–9 then 0 for the first ten periods; later periods repeat 1, 2, 3... in thinkorswim's own colours. Session times use thinkorswim's US Eastern clock and follow US daylight saving automatically. On a Sydney-time chart the profiles simply sit at different axis times when either country changes its clocks.

3. Inputs

Input	Default	What it does
rowHeight	5.0	Price range per row. /NQ 5, /ES 2. Must be set per chart.
numberOfProfiles	10	Profiles drawn (about 5 trading days).
playgroundPercent	70	TPO value area %.
volumeValueAreaPercent	70	Volume value area %.
showInitialBalance / initialBalancePeriods	yes / 2	Initial Balance bracket over the first hour.
profileOpacity	75	Strength of the whole profile. Lower to ~65 if value areas feel heavy.
showRTHSeparator	yes	Faint green dashed line at 09:30.
showRTHBackground	yes	Faint background tint during RTH.
showRowHeightLabel	no	Yes shows the row height in use (for checking).

4. Colour key

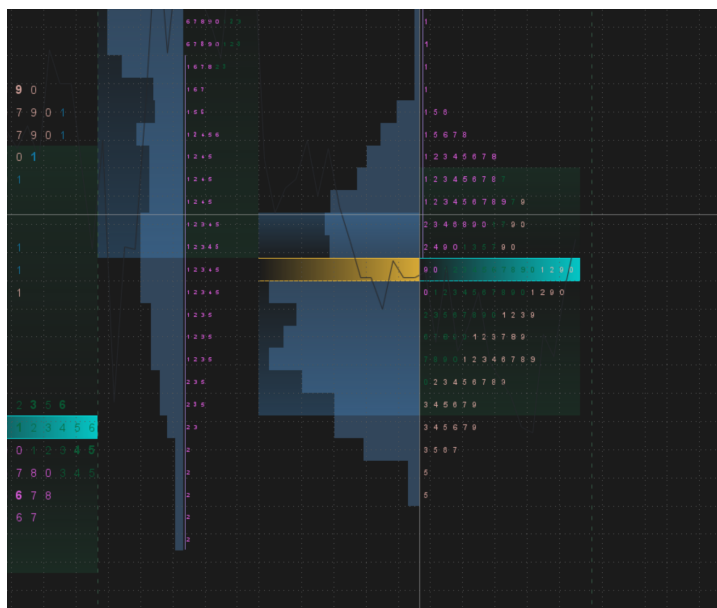
	Element	Meaning
	TPO digits	A 30-minute period traded at that price row.
	Green shading	TPO value area (70% of TPOs).
	Cyan bar	TPO point of control.
	Bright blue histogram	Volume value area (70% of volume).
	Dark slate histogram	Volume outside the value area.
	Gold bar	Volume point of control.
	Lavender bracket	Initial Balance (first hour).
	Green dashed line	RTH open (09:30).
	Background tint	RTH session.

All colours can be changed under **Study settings** → **Globals** without editing code.

5. Reading the digits

There is no font-size setting: each digit is sized to fit its cell, which is **one row tall by one 30-minute bar wide**. If digits look like dotted lines, the cells are too small on screen:

- **Zoom in on the time axis** (magnifier or drag the axis) — the biggest effect. About 2–3 days on screen works well.
- Give the chart more width in the Flexible Grid, or maximise it.
- Raise rowHeight (e.g. 10 on /NQ) only if rows are too short; this makes the profile coarser.



Example: volume histogram on the left of each profile, TPO digits on the right, gold volume POC and cyan TPO POC.

6. Known limitations

- RTH and Globex profiles share one colour palette (one MonkeyBars object = one palette); the RTH tint marks the session instead.
- The RTH profile includes 16:00–17:00 post-close trade. Usually negligible; can stretch the profile on after-hours earnings days.
- Row height cannot be chosen automatically by symbol or price; thinkorswim requires a fixed value. Set it once per chart.
- Volume histogram always sits to the left of the digits within each profile; this order cannot be changed.
- Designed for 30-minute charts; other aggregations will number periods incorrectly.

7. Troubleshooting

Symptom	Fix
Digits appear as dotted lines	Zoom in on the time axis; check rowHeight (turn on showRowHeightLabel).
Profiles look very spiky / tall on /NQ	rowHeight set too low (e.g. 1 or 2). Use 5.
Old colours or settings still showing	Remove and re-add the study; check the latest code was saved.
No Globex profiles	Enable extended-hours trading on the chart.